Atom

- They’re at their simplest form.
- Contain a nucleus, protons, neutrons and electrons.
- 8 is a magic number and makes them stable.
- Protons are positive and electrons are negative.

Ion

- An atom can give or receive an electron to become an ion.
- It can be either a positive or a negative charged atom.
- It has an atomic weight.
- It is an element on the periodic table.
- Can also be a combination of elements.
- Needs to be balanced.

Molecule

- 2 or more chemically bonded atoms.
- Small, found everywhere.

Element

- Listed on the periodic table.
- Can be solid, liquid and gas.
- Contain protons, neutrons and electrons.
- Have atomic mass which is the sum of protons and neutrons.
- Contain all the same kinds of atoms.
- Can’t be chemically broken down.

Compound

- Group of molecules.
- Can form chemicals such as carbon dioxide, water, nitrous oxide.

Mixture

- When 2 or more substances are not chemically combined but are found together.
- Go through different processes.
- Different from a compound.
- For example air, soil, cake mixture.